Fundamentals of Deep Learning

Lecture Notes

Muxammadrizo Maribjonov

November 3, 2026

Contents

1 From Logistic Regression to MLPs **2**

1.1 Recap of Logistic Regression . . . 2
1.2 Cross Entropy, KL Divergence, and Maximum Likelihood . . . 2
1.3 Logistic Regression as a Neural Network: The Perceptron . . . 2
1.4 Multilayer Perceptrons . . . 2
1.5 Matrix Dimensions . . . 2
1.6 Other Activation Functions and Loss Functions . . . 2
1.7 Representational Power of MLPs . . . 2
1.8 Summary, Further Reading, References . . . 2

2 MLPs and Backpropagation **3**

2.1 Recap of Multilayer Perceptrons . . . 3
2.2 Recap of Chain Rule of Calculus . . . 3
2.3 Calculating Gradients with Backpropagation . . . 3
2.4 Backprop for a Linear Layer in Matrix Form . . . 3
2.5 Full Matrix Form of Backpropagation Equations . . . 3
2.6 Summary, Further Reading, References . . . 3

Chapter 1

From Logistic Regression to MLPs

1.1 Recap of Logistic Regression

1.2 Cross Entropy, KL Divergence, and Maximum Likelihood

1.3 Logistic Regression as a Neural Network: The Perceptron

1.4 Multilayer Perceptrons

1.5 Matrix Dimensions

1.6 Other Activation Functions and Loss Functions

1.7 Representational Power of MLPs

1.8 Summary, Further Reading, References

Chapter 2

MLPs and Backpropagation

2.1 Recap of Multilayer Perceptrons

2.2 Recap of Chain Rule of Calculus

2.3 Calculating Gradients with Backpropagation

2.4 Backprop for a Linear Layer in Matrix Form

2.5 Full Matrix Form of Backpropagation Equations

2.6 Summary, Further Reading, References